4.10.24

EE25BTECH11022 - sankeerthan

Question:Find the equations of the lines through the point of intersection of the line $x - y + 1 = 0$ and $2x - 3y + 5 = 0$ and whose distance from the point $(3, 2)$ is $\frac{7}{5}$.

Solution: Given lines are

$$\begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 1 = 0, \begin{pmatrix} 2 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 5 = 0 \tag{1}$$

$$\begin{pmatrix} 1 & -1 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ -5 \end{pmatrix} \tag{2}$$

Write as augmented matrix

$$\left(\begin{array}{cc|c} 1 & -1 & -1 \\ 2 & -3 & -5 \end{array}\right) \xrightarrow{R_2 \to R_2 - 2R_1} \left(\begin{array}{cc|c} 1 & -1 & -1 \\ 0 & -1 & -3 \end{array}\right) \tag{3}$$

From second row: $-y = -3 \implies y = 3$
Substitute in first row: $x - 3 = -1 \implies x = 2$
Intersection point of given lines is $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$
General line through (x_0, y_0) with normal vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is

$$\begin{pmatrix} a & b \end{pmatrix} \left(\begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right) = 0 \tag{4}$$

Distance from point $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ to line is

$$Distance = \frac{\left| \begin{pmatrix} a & b \end{pmatrix} \left(\begin{pmatrix} 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right) \right|}{\sqrt{\begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}}} = \frac{7}{5} \tag{5}$$

$$\frac{\left| \begin{pmatrix} a & b \end{pmatrix} \left(\begin{pmatrix} 3-2 \\ 2-3 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right) \right|}{\sqrt{\begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}}} = \frac{7}{5} \tag{6}$$

$$\frac{|a - b|}{\sqrt{a^2 + b^2}} = \frac{7}{5} \tag{7}$$

Let t = $\frac{a}{b}$ $(b \neq 0)$

$$\frac{|t-1|}{\sqrt{t^2+1}} = \frac{7}{5} \tag{8}$$

Squaring on both sides

$$25(t-1)^2 = 49(t^2+1) \tag{9}$$
$$25(t^2-2t+1) = 49t^2+49 \tag{10}$$
$$25t^2-50t+25 = 49t^2+49 \tag{11}$$
$$24t^2+50t+24 = 0 \tag{12}$$
$$12t^2+25t+12 = 0 \tag{13}$$
$$t = \frac{-25 \pm \sqrt{25^2-4\times12\times12}}{2\times12} = \frac{-25\pm7}{24} \tag{14}$$

$t_1 = -\frac{3}{4}, t_2 = -1$
Normal vectors $(let b = 1)$:

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \frac{-3}{4} \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \tag{15}$$

Equation constants c = -a(2) - b(3)

$$c_1 = (-2)\times\left(-\frac{3}{4}\right) - 3\times1 = \frac{3}{2} - 3 = \frac{-3}{2} \tag{16}$$
$$c_2 = (-2)\times(-1) - 3\times1 = 2-3 = -1 \tag{17}$$

required equations of line are

$$\begin{pmatrix} \frac{-3}{4} & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} - \frac{3}{2} = 0 \tag{18}$$
$$\begin{pmatrix} -1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} - 1 = 0 \tag{19}$$

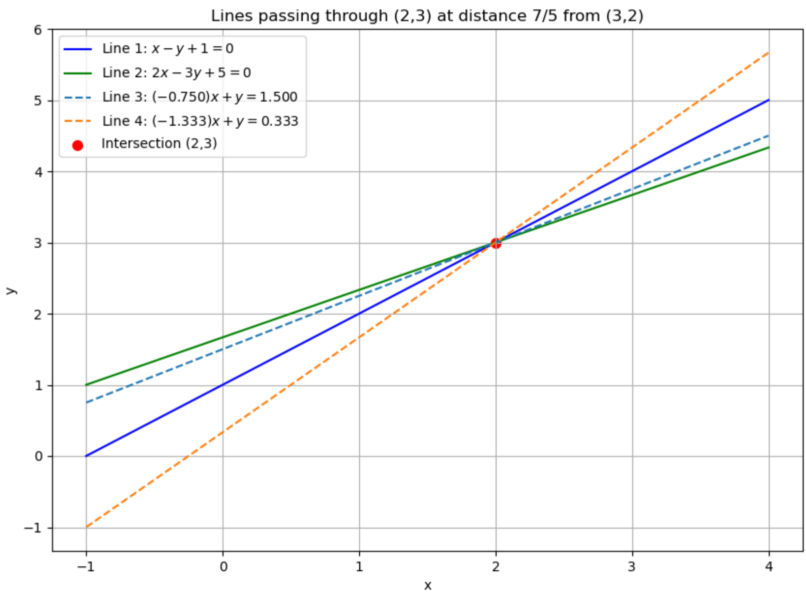
Lines passing through (2,3) at distance 7/5 from (3,2)
Line 1: $x - y + 1 = 0$
Line 2: $2x - 3y + 5 = 0$
Line 3: $(-0.750)x + y = 1.500$
Line 4: $(-1.333)x + y = 0.333$
Intersection (2,3)
6
5
4
3
2
1
0
−1
y
−1
0
1
2
3
4
x

Fig. 0.1